HOMEWORK 9

1. A finite field extension K/F is called *purely inseparable* if every element in K that is separable over F is contained in F. Let K/F be a purely inseparable field extension and $char(F) = p > 0$. Prove that there is an integer $n > 0$ such that $x^{p^n} \in F$ for every $x \in K$.

2. Let E/F be a normal field extension and $G = \mathrm{Gal}(E/F)$. Prove that the extension E^G/F is purely inseparable.

3. Let K/F be a finite field extension and let L/F be the maximal separable subextension in K/F. Prove that the extension K/L is purely inseparable.

4. Let K/F be an algebraic field extension (not necessarily finite). Prove that every field homomorphism $K \to K$ over F is an isomorphism.

~~5.~~ Let K/F be an algebraic field extension (not necessarily finite) and $\sigma : K \to K$ a field isomorphism over F (not necessarily of finite order). Let K^σ be the subfield of all σ-invariant elements in K. Prove that if L is a subfield of K containing K^σ and finite over K^σ, then L/K^σ is a cyclic Galois field extension.

6. Prove that an algebraic (not necessarily finite) extension of a perfect field is perfect.

7. Let K/F be a finite field extension. For every $x \in K$ write l_x for the linear operator $K \to K$ over F taking y to xy. Let $\mathrm{Tr}_{\mathrm{K/F}}(x)$ and $\mathrm{N}_{K/F}(x)$ the trace and the determinant of l_x respectively.

a) Prove that $\mathrm{Tr}_{K/F}(x+x') = \mathrm{Tr}_{K/F}(x)+\mathrm{Tr}_{K/F}(x')$ and $\mathrm{N}_{K/F}(xx') = \mathrm{N}_{K/F}(x)\cdot \mathrm{N}_{K/F}(x')$ for all $x, x' \in K$.

b) Compute the trace and the norm of $a + b\sqrt{2}$ for the extension $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ (here $a, b \in \mathbb{Q}$).

8. Let $L/K/F$ be a tower of finite field extensions. Prove that $\mathrm{Tr}_{L/F} = \mathrm{Tr}_{\mathrm{K/F}} \circ \mathrm{Tr}_{L/K}$ and $\mathrm{N}_{L/F} = \mathrm{N}_{K/F} \circ \mathrm{N}_{L/K}$.

9. Let $p_1, p_2, \ldots, p_n$ be distinct prime integers. Prove that

$$\sqrt{p_1} + \sqrt{p_2} + \cdots + \sqrt{p_n} \notin \mathbb{Q}.$$

(Hint: Use the trace map.)

10. Let K/F be a Galois field extension with the Galois group G. Let H be a subgroup of G. Prove that $\mathrm{Gal}(K^H/K)$ is isomorphic to the factor group $N_G(H)/H$, where $N_G(H)$ is the normalizer of H in G.